Outline

- Inter-observational Distances
- Partition-based Clustering
- Hierarchical Clustering
- Diagnostics and Optimizing Number of Clusters
- Density-based Clustering

Partition-based Clustering

Basic Idea: Specify the number of clusters into which the data will be partitioned, and then perform computation to group data so that

1. observations within clusters are similar (low distances), and
2. observations in different clusters are dissimilar (high distances).

We will address the optimal number of clusters separately.

K-Means Clustering

Let $C_1, \ldots, C_K$ denote sets containing the indices of observations within the K clusters such that

1. $C_1 \cup C_2 \cup \cdots \cup C_K = \{1, 2, \ldots, n\}$, and
2. $C_k \cap C_l = \emptyset$ for $k \neq l$.

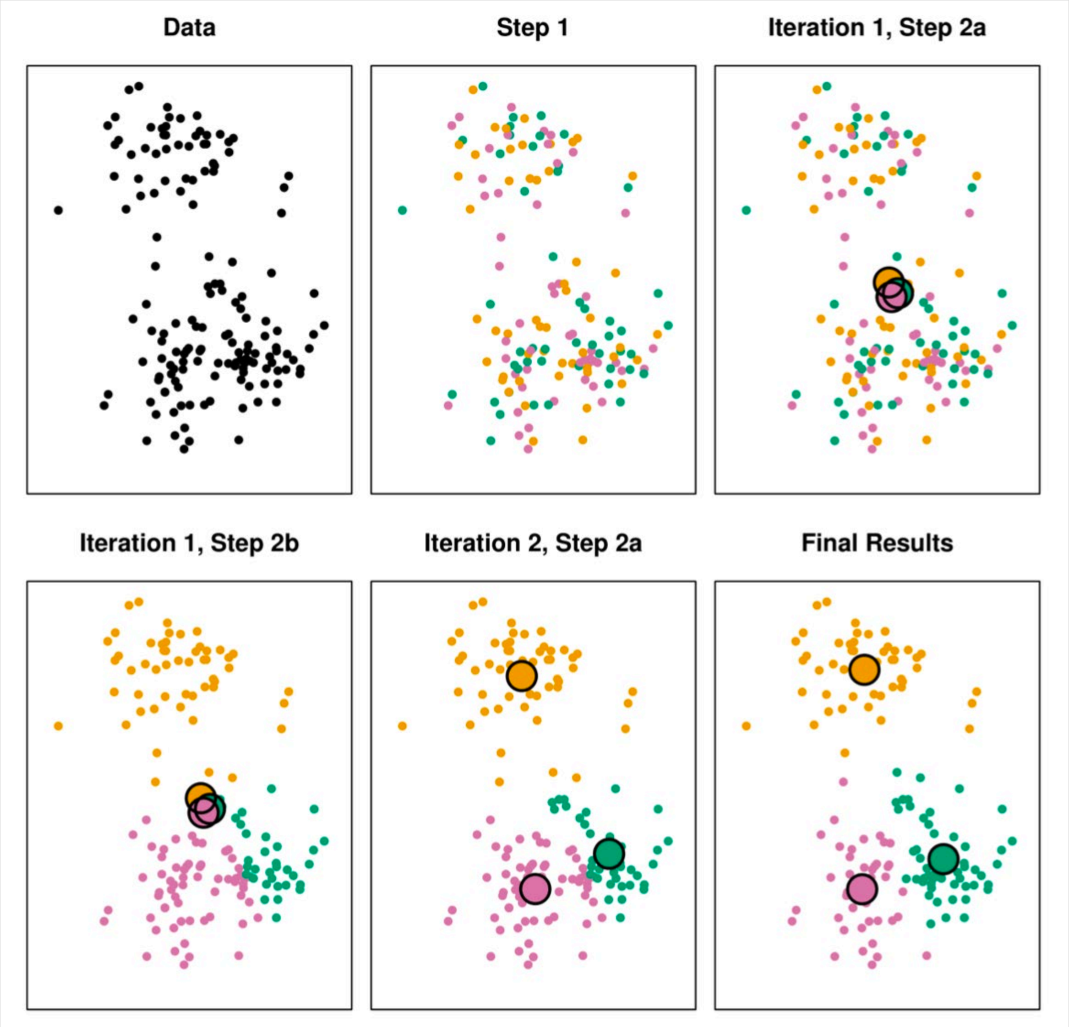
Let $W(C_K) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \sum_{j=1}^{p} \left(X_{ij} - X_{i'j}\right)^2$ be the within cluster variation.

The *goal of K-Means clustering* is to find $C_1, \ldots, C_K$ such that $\sum_{k=1}^{K} W(C_k)$ is minimized.

K-Means Clustering Algorithm

1. Randomly assign each observation to one of K clusters at random.
2. Repeat the following steps until clusters do not change:
    1. For each cluster k, compute the cluster centroid $\bar{x}_k$ (the variable-wise average of the observations in cluster k).
    2. Given the k centroids, reassign all observations to clusters based on their closeness to the centroids.

K-Means Clustering Algorithm



K-Means Clustering Algorithm

- Requires you to select K in advance.
- The algorithm is *locally optimal*, not globally optimal.
  - i.e., you can get different cluster results depending on the initial cluster assignments.

- Potential Solutions
  - Try different values of K and compare the results.
  - Try different initial cluster assignments in parallel and chose the one with the best within-cluster sum of squared deviations.

K-Means Clustering Algorithm

- Six results from different initializations

Example: Violent Crimes by State

First few observations rescaled (subtract mean, divide by standard deviation)

	Murder	Assault	UrbanPop	Rape
State				
Hawaii	-0.58	-1.51	1.22	-0.11
Indiana	-0.14	-0.70	-0.04	-0.03
New Mexico	0.84	1.38	0.31	1.17
Washington	-0.88	-0.31	0.52	0.54
Maine	-1.32	-1.06	-1.01	-1.45
Alabama	1.26	0.79	-0.53	-0.00

Example: Violent Crimes by State

We chose 4 clusters for now

State	Murder	Assault	UrbanPop	Rape	Cluster
Alabama	1.255179	0.790787	-0.526195	-0.003451	1
Alaska	0.513019	1.118060	-1.224067	2.509424	1
Arizona	0.072361	1.493817	1.009122	1.053466	1
Arkansas	0.234708	0.233212	-1.084492	-0.186794	2
California	0.281093	1.275635	1.776781	2.088814	1

Clustering Diagnostics: Silhouette Plot

Once a clustering has been determined, let

- a_i = average dissimilarity between observation i and the other points in the cluster to which i belongs
- b_i = average dissimilarity between observation i and the other points in the *next closest* cluster to observation i.

We define $s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$ to be the silhouette score for observation i.

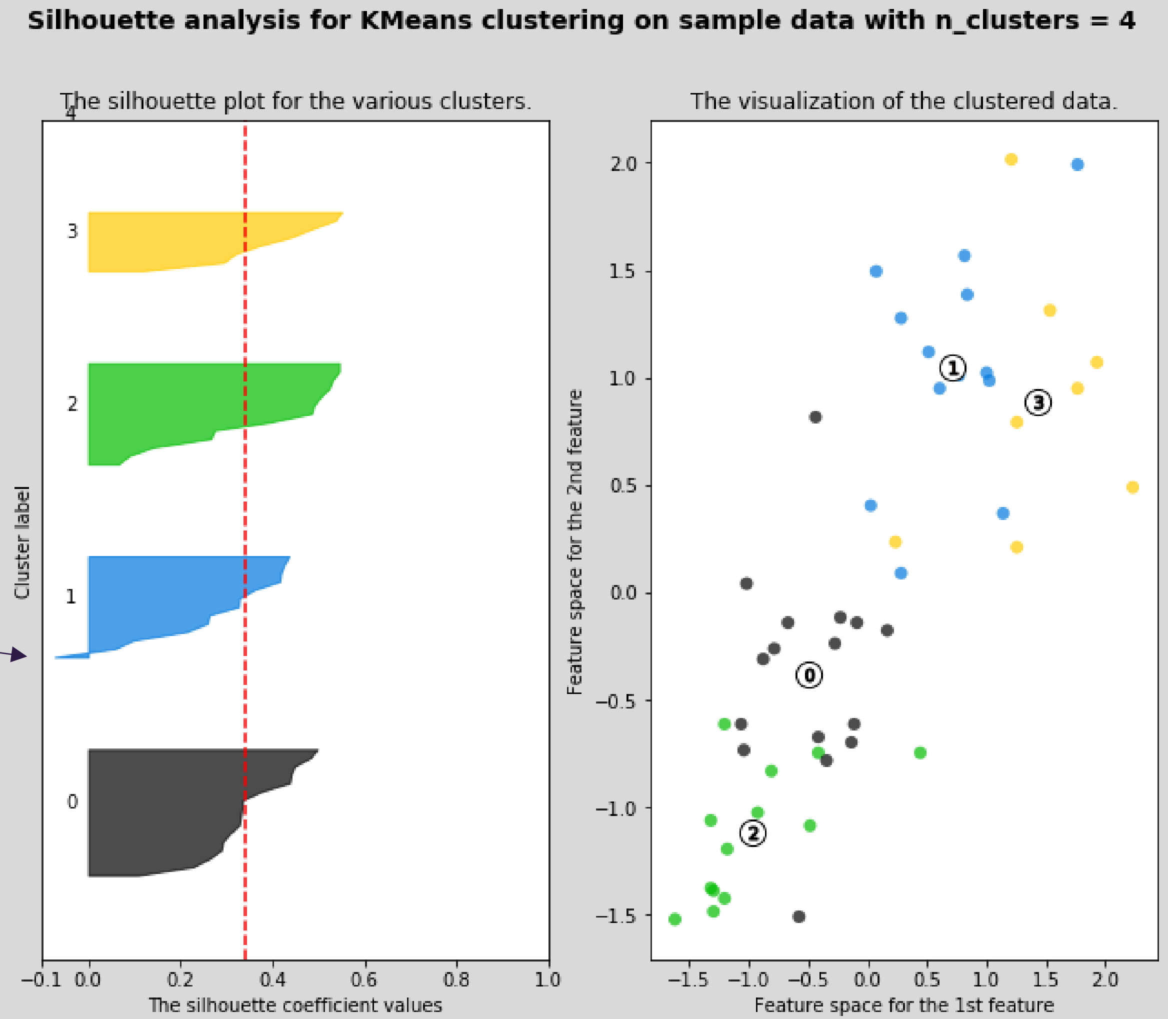
Clustering Diagnostics: Silhouette Plot

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

- Interpretation
  - Observations with $s_i \approx 1$ are well-clustered
  - Observations with $s_i \approx 0$ lie between two clusters
  - Observations with $s_i < 0$ are probably in the wrong cluster

Example: Violent Crimes by State

Something may be in the wrong cluster



Example: Violent Crimes by State

	State	Murder	Assault	UrbanPop	Rape	StateAbbrv
24	Missouri	9.0	178	70	28.2	MO